ON-BOARD DIAGNOSTIC [HEADLIGHT AUTO LEVELING SYSTEM]

id091800905500

Outline

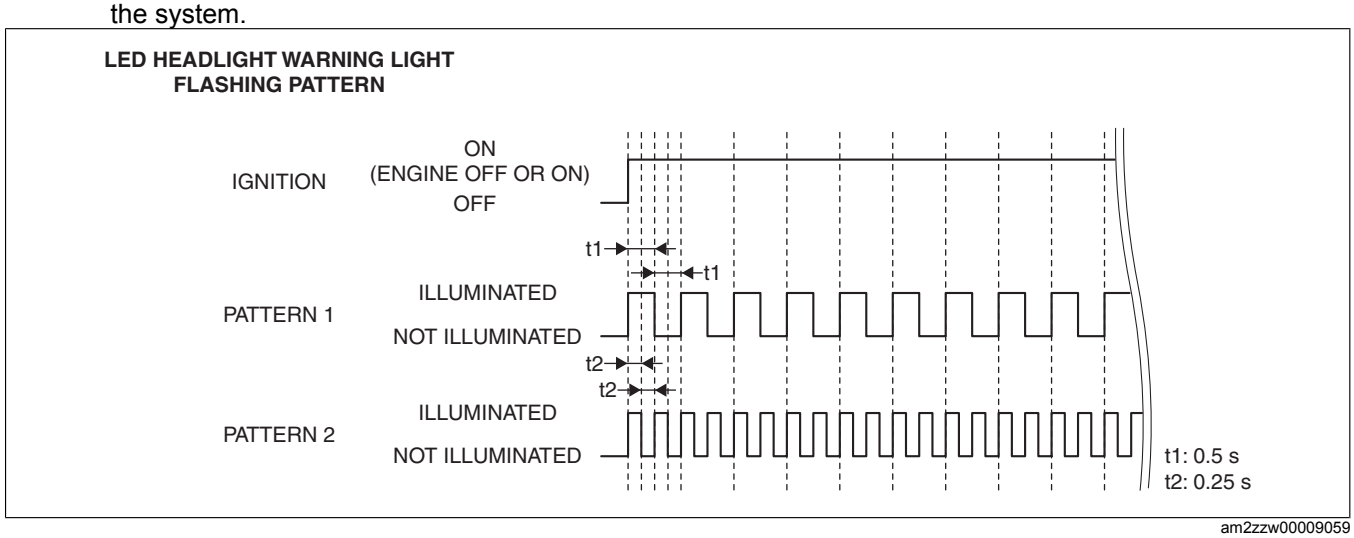
- The on-board diagnostic function consists of the following functions.
  - — A malfunction detection function, which detects overall malfunctions in the auto leveling control module-related parts.
  - — A memory function, which stores detected DTCs.
  - — A display function, which indicates malfunction locations and status via DTC output.
  - — A PID/data monitor function, which reads out specific input/output signals and verifies the input/output condition.
- Using the Mazda Modular Diagnostic System (M-MDS), DTCs can be read out and deleted, and the PID/data monitor function can be activated.

Malfunction Detection Function

- Detects malfunctions in input/output signals.
- If a malfunction occurs, the auto leveling control module records the malfunction as a DTC. A recorded DTC can be read by the Mazda Modular Diagnostic System (M-MDS).

Note

- Auto leveling control module flashes the LED headlight warning light as follows if there is a malfunction in the system.



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DTC table

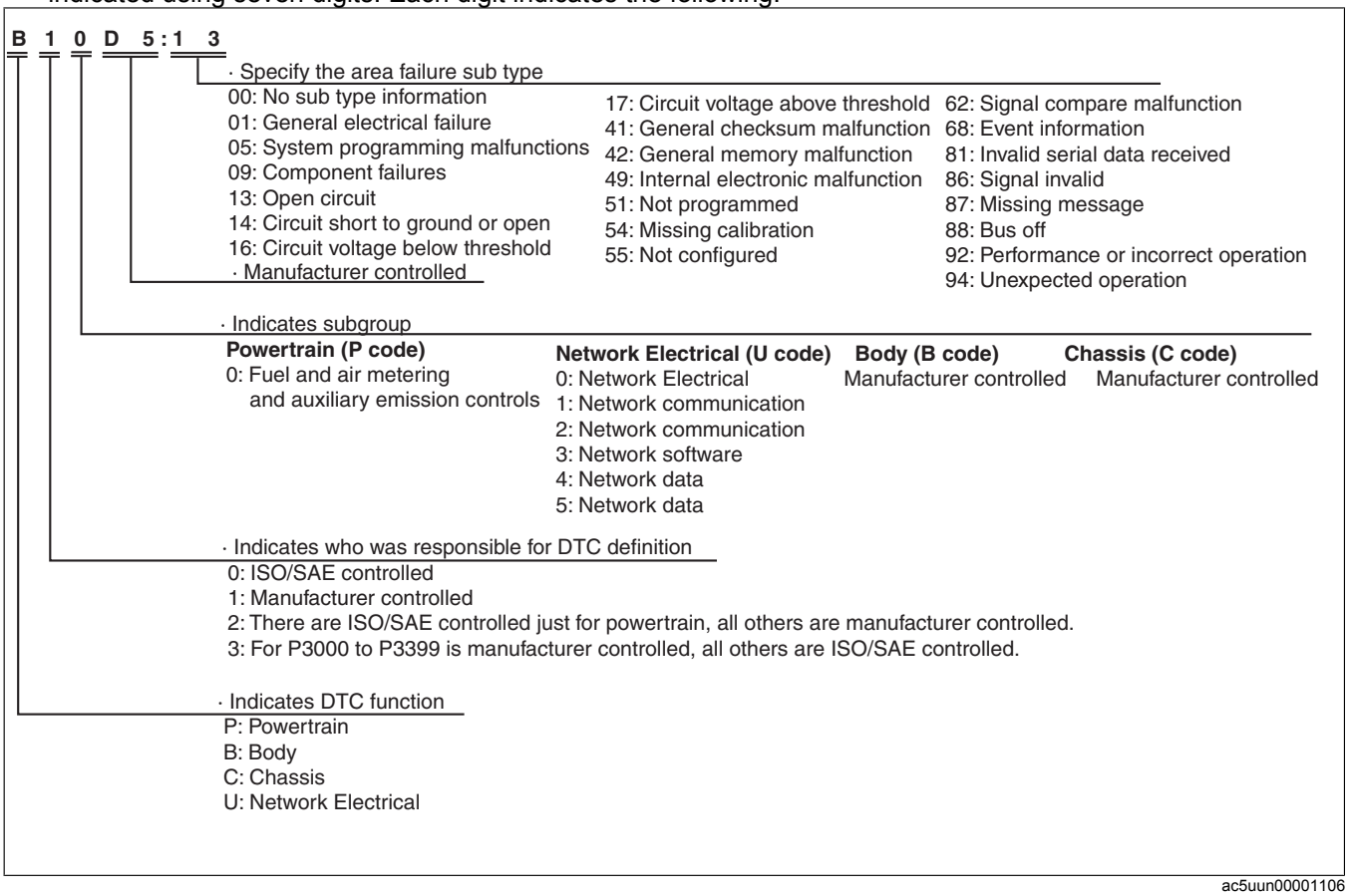
×: Applicable
—: Not applicable

DTC No.	LED headlight warning light	Description	Fail-safe	Drive cycle	Self test type*1	Memory function
B1041:54	Flash (Pattern 2)	Headlight auto leveling system initialization error	×	—	—	×
B1044:01	Flash (Pattern 1)	Auto leveling sensor circuit malfunction	×	—	—	×
B1D00:13	Flash (Pattern 1)	Headlight (LH) circuit malfunction	×	—	C	×
B1D01:13	Flash (Pattern 1)	Headlight (RH) circuit malfunction	×	—	C	×
U0001:88	Flash (Pattern 1)	Unit communication error (HS-CAN)	×	—	C	×
U0100:00	Flash (Pattern 1)	Communication error with PCM	×	—	C	×
U0140:00	Flash (Pattern 1)	Communication error with BCM	×	—	C	×
U0155:00	Flash (Pattern 1)	Communication error with instrument cluster	×	—	C	×
U0423:68	—	Error signal received from instrument cluster	×	—	C	—

DTC No.	LED headlight warning light	Description	Fail-safe	Drive cycle	Self test type*1	Memory function
U2005:86	Flash (Pattern 1)	Error signal received from PCM	×	—	C	×
U2300:54	Flash (Pattern 1)	Error configuration data received from instrument cluster	×	—	C	×
U2300:55	Flash (Pattern 1)	Instrument cluster configuration not implemented	×	—	C	×
U2300:56	Flash (Pattern 1)	Configuration data unmatched with instrument cluster	×	—	C	×
U3000:42	Flash (Pattern 1)	Auto leveling control module internal malfunction	×	—	C	×
U3000:49	Flash (Pattern 1)	Auto leveling control module internal malfunction	×	—	C	—
U3003:16	—	Auto leveling control module low power supply voltage input	×	—	C	—
U3003:17	—	Auto leveling control module high power supply voltage input	×	—	C	—

DTC 7-digit code definition

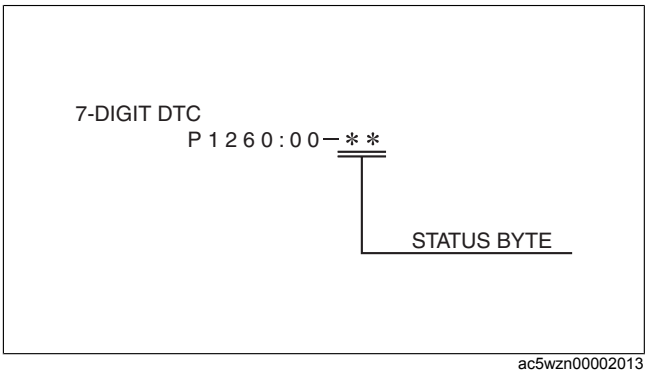
- When related systems or components have failed, the CM stores the DTC of the malfunctioning part in the CM memory, and allows for the retrieval of the store data using scanning tool when necessary. The DTCs are indicated using seven digits. Each digit indicates the following.



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Status byte for DTC

- The two digits (two digits after hyphen (-)) after the 7-digit DTC.
- A code which indicates the pending code, current/ past malfunction status, or warning illumination status.
- Can be read by performing a CMDTC self-test using the Mazda Modular Diagnostic System (M-MDS).
- For details on the status byte, refer to the explanation on the Mazda Modular Diagnostic System (M-MDS) when reading the DTC.



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Detection condition for the applicable DTC

DTC No.	Description	Detection condition
B1041:54	Headlight auto leveling system initialization error	• Headlight auto leveling system initialization is not performed with the ignition switched ON (engine off or on).
B1044:01	Auto leveling sensor circuit malfunction	• Auto leveling control module detected the auto leveling sensor circuit voltage of **0.25 V or less** or **4.75 V or more** for **10 s or more** with the ignition switched ON (engine off or on).
B1D00:13	Headlight (LH) circuit malfunction	• The auto leveling control module detected an open circuit in the headlight (LH) circuit during the headlight LO on control with the ignition switched ON (engine off or on).
B1D01:13	Headlight (RH) circuit malfunction	• The auto leveling control module detected an open circuit in the headlight (RH) circuit during the headlight LO on control with the ignition switched ON (engine off or on).
U0001:88	Unit communication error (HS-CAN)	• Auto leveling control module detected the CAN bus communication line (HS-CAN) malfunction **10 times** continuously.
U0100:00	Communication error with PCM	• Auto leveling control module could not receive the CAN signal from the PCM for **5 s or more**.
U0140:00	Communication error with BCM	• Auto leveling control module could not receive the CAN signal from the BCM for **5 s or more**.
U0155:00	Communication error with instrument cluster	• Auto leveling control module could not receive the CAN signal from the instrument cluster for **5 s or more**.
U0423:68	Error signal received from instrument cluster	• Auto leveling control module received headlight illumination state signal which undefined for **5 s or more** with the ignition switched ON (engine off or on).
U2005:86	Error signal received from PCM	• Auto leveling control module received the vehicle speed signal error from the PCM for **5 s or more** with the ignition switched ON (engine off or on).
U2300:54	Error configuration data received from instrument cluster	• Auto leveling control module received a signal which indicates the instrument cluster configuration is not performed.
U2300:55	Instrument cluster configuration not implemented	• Auto leveling control module received a signal which indicates the instrument cluster configuration is not performed.
U2300:56	Configuration data unmatched with instrument cluster	• Configuration data of the auto leveling control module and instrument cluster are not matched.
U3000:42	Auto leveling control module internal malfunction	• Auto leveling control module detects the internal EEPROM malfunction.
U3000:49	Auto leveling control module internal malfunction	• Watchdog timer overflow inside the auto leveling control module detected.
U3003:16	Auto leveling control module low power supply voltage input	• Auto leveling control module power supply circuit voltage of **9.0 V or less** is detected for **5 s or more** with the ignition switched ON (engine off or on).
U3003:17	Auto leveling control module high power supply voltage input	• Auto leveling control module power supply circuit voltage of **18.1 V or more** is detected for **5 s or more** with the ignition switched ON (engine off or on).

Snapshot data

Note
- The auto leveling control module stores the following two types of snapshot data (vehicle information) when a DTC is detected and displays them in the Mazda Modular Diagnostic System (M-MDS).
  - — Vehicle information detected by auto leveling control module
  - — Vehicle information detected by instrument cluster and received by auto leveling control module via CAN communication

- The data for all DTCs currently detected is stored.

—: Not applicable

Snapshot data item	Unit		Data contents	Data read/use method	Corresponding data monitor items
AAT	°C	°F	Ambient temperature	—	—
APP_STATUS	Accelerator Pedal Off/Under20%/ Over20%/FAIL		Accelerator pedal position status	—	—
CFG_STATUS	Config Complete/ Not Configured/ Config Error		Instrument cluster configuration status	—	—
ECT_STATUS	Under 0 degrees C/ 0 - Under 80 degrees C/Over 80 degrees C/FAIL		Engine coolant temperature status	—	—
IC_VPWR	V		Instrument cluster power supply voltage	• The auto leveling control module constantly receives the power supply voltage value of the instrument cluster sent via CAN communication from the instrument cluster. • If a DTC is detected, the auto leveling control module records the power supply voltage of the instrument cluster when the DTC was detected, and it is displayed in the Mazda Modular Diagnostic System (M-MDS).	VPWR*1
IG-ON_TIMER	hh:mm:ss*2		Elapsed time since ignition was switched ON (engine off or on) **Note** • The instrument cluster records the elapsed time since the ignition was switched ON (engine off or on).	• The auto leveling control module constantly receives the elapsed time since the ignition was switched ON (engine off or on) sent via CAN communication from the instrument cluster. • If a DTC is detected, the auto leveling control module records the elapsed time since the ignition was switched ON (engine off or on) when the DTC was detected, and it is displayed in the Mazda Modular Diagnostic System (M-MDS).	—

Snapshot data item	Unit		Data contents	Data read/use method	Corresponding data monitor items
PWR_MODE_KEY	Key Out/Key Recently Out (Position 0)/ Accessory (Position 1)/Post Ignition (Position 2)/Ignition On (Position 2)/ Running (Position 2)/Running - Starting		• Key Out: Ignition switched off • Key Recently Out (Position 0): Elapsed time **within 3 s** since ignition was switched off • Accessory (Position 1): Ignition is switched to ACC • Post Ignition (Position 2): Elapsed time **within 3 s** since ignition was switched ON (engine off or on) • Ignition On (Position 2): Ignition switched ON (engine off) • Running (Position 2): Ignition switched ON (engine on) • Running - Starting: Cranking condition	• The auto leveling control module constantly receives the ignition switch status sent via CAN communication from the instrument cluster. • If a DTC is detected, the auto leveling control module records the ignition switch status when the DTC was detected, and it is displayed in the Mazda Modular Diagnostic System (M-MDS).	—
RPM_STATUS	Engine Stop/ Under1500rpm/ Over1500rpm/FAIL		Engine speed status	• The auto leveling control module constantly receives the engine speed sent via CAN communication from the instrument cluster. • If a DTC is detected, the auto leveling control module records the engine speed when the DTC was detected, and it is displayed in the Mazda Modular Diagnostic System (M-MDS).	TACHOMTR*1
SHIFT_STATUS	P/N/D/R/FAIL		Selector lever position status	• The auto leveling control module constantly receives the selector lever position sent via CAN communication from the instrument cluster. • If a DTC is detected, the auto leveling control module records the selector lever position when the DTC was detected, and it is displayed in the Mazda Modular Diagnostic System (M-MDS).	—
TOTAL_DIST	km	Miles	Accumulated total traveled distance from completion of vehicle until auto leveling control module detects DTC (Odometer value in instrument cluster)	The total traveled distance from which the auto leveling control module detects DTCs to the present can be calculated by performing the following procedure. 1. Verify the odometer value in the instrument cluster. 2. Verify the snapshot data item TOTAL_DIST. 3. Subtract 2 from 1.	—

Snapshot data item	Unit	Data contents	Data read/use method	Corresponding data monitor items
TOTAL_TIME	hh:mm:ss*2	Accumulated total elapsed time since vehicle completion until auto leveling control module detects a DTC **Note** • When the ROOM fuse is removed, and the ignition is switched off, the time is not included in the elapsed time.	The elapsed time from which the auto leveling control module detects DTCs to the present can be calculated by performing the following procedure. 1. Verify the instrument cluster PID item TOTAL_TIME. 2. Verify the snapshot data item TOTAL_TIME. 3. Subtract 2 from 1.	TOTAL_TIME*1
VPWR	V	Auto leveling control module power supply voltage	—	VPWR_IG
VSPD_STATUS	Stop/0-10km/h/ Over10km/h/FAIL	Vehicle speed status	• The auto leveling control module constantly receives the vehicle speed sent via CAN communication from the instrument cluster. • If a DTC is detected, the auto leveling control module records the vehicle speed when the DTC was detected, and it is displayed in the Mazda Modular Diagnostic System (M-MDS).	SPEEDOMTR*1

*1 : Instrument cluster PID (See ON-BOARD DIAGNOSTIC [INSTRUMENT CLUSTER].)

*2 : The seconds may be indicated after the decimal point.

PID/Data Monitor Function

- With the PID/data monitor function, input/output signal monitor items set in the auto leveling control module can be selected and read out in real-time.
- The items which can select are as follows:

PID	Unit/ Operation	Data contents	Module control terminal
H/L_BO_L	OK/NG	Displays the headlight LO (LH) open circuit detection status. (Detects during low beam on)	I
H/L_BO_R	OK/NG	Displays the headlight LO (RH) open circuit detection status. (Detects during low beam on)	Z
R_HGT_S	V	Displays voltage at auto leveling sensor	F, J, T
R_HGT_S_INI	V	Displays voltage at headlight auto leveling system initial setting	F, J, T
VPWR_IG	V	Displays auto leveling control module power supply voltage	A, C
VSPD	KPH, MPH	Displays vehicle speed	AB, AD